Multiple Reactions in a Non-isothermal PFR Python solution

The gas phase elementary reactions 2A ⇋ B and 2B + A --> C are carried out adiabatically in a PFR with no pressure drop. Plot the temperature and concentrations along the volume of the PFR.

Info:

k_{1A} = 50 M^{-1} min^{-1} at 350 K with E = 8 kJ/mol
k_{2C} = 4000 M^{-2} min^{-1} at 310 K with E = 4 kJ/mol
$K_{C,A}$ = 10 M^{-1} at 315 K
ΔH_{1A}^{rxn} = -20 kJ/mol A
ΔH_{2B}^{rxn} = +30 kJ/mol B
V = 1 L
T(0) = 150C
CA0 = 0.1 M
FA0 = 1 mol/min
CPA = CPB = 0.09 kJ/mol*K
CPC = 0.18 kJ/mol*K

Decision Guide: Multiple Reactions in a Non-Isothermal PFR

Goal

Develop a mathematical model that predicts:

- Temperature as a function of reactor volume, $T(V)$
- Concentration profiles for all species
- How reaction rates change throughout the reactor
- The interaction between reaction kinetics and the energy balance

Step 1: Identify the Type of Problem

Ask yourself:

What reactor type is this?

→ PFR

Is it isothermal?

→ No

Is it adiabatic?

→ Yes

Multiple reactions?

→ Yes

Pressure drop?

→ No

Because this is an adiabatic, non-isothermal PFR with multiple reactions, you will need:

- A mole balance for each species
- An energy balance
- Temperature-dependent rate constants

These equations must be solved simultaneously.

Step 2: Identify All Species

List every species appearing in the reaction network: A, B, C

These become the dependent variables in the model.

We will eventually solve for:

FA(V), FB(V), FC(V), T(V)

Step 3: Identify the Reactions

Reaction 1: 2A ⇌ B
Reaction 2: 2B + A → C

Notice that:

- Reaction 1 is reversible.
- Reaction 2 is irreversible.
- Species A and B participate in both reactions.

Step 4: Write the Individual Reaction Rates

Treat the reactions separately.

Reaction 1

Write forward and reverse rates using:

- $k_{1,f}$
- $K_{C,1}$

For a reversible reaction: Net Rate = Forward Rate − Reverse Rate

Reaction 2

Use the elementary-rate expression given in the problem statement.

Do not combine reactions at this stage.

Step 5: Calculate Temperature-Dependent Rate Constants

The problem provides:

- Reference rate constants
- Activation energies
- Reference temperatures

Use the Arrhenius equation:

$$k(T) = k(T_{ref}) \exp\left[\frac{E}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right]$$

CBE 3610 Reactor Design Handbook Equation (49)

Compute:

k1(T)

k2(T)

inside the ODE function.

Step 6: Write Species Mole Balances

For a PFR:

$$\frac{dF_i}{dV} = r_i$$

CBE 3610 Reactor Design Handbook Equation (18) for a general species balance or Equation (16) for PFR design.

Write one balance for each species:

Species A

Include contributions from:

- Reaction 1
- Reaction 2

Species B

Include:

- Production and consumption by Reaction 1
- Consumption by Reaction 2

Species C

Include:

- Production by Reaction 2 only

Step 7: Convert Flowrates to Concentrations

Reaction rates require concentrations.

Use:

Ci = Fi / v

Because:

- Gas phase
- Change in the total number of moles
- Variable temperature

the volumetric flowrate changes along the reactor. Calculate concentrations from the current values of:

FA, FB, FC, T

at each integration step.

Step 8: Build the Heat-Generation Term

The energy balance must include contributions from **both reactions**. Total heat generation is:

$$\dot{Q}_{gen} = r_1 \Delta H_{rxn,1} + r_2 \Delta H_{rxn,2}$$

For this problem:

Reaction 1: Exothermic
Reaction 2: Endothermic

Both reactions influence the reactor temperature simultaneously.

Step 9: Write the Adiabatic Energy Balance

Because the reactor is adiabatic: $Q = 0$

Use the PFR energy balance framework. For multiple reactions, the temperature changes because of:

Heat released by Reaction 1 + Heat absorbed by Reaction 2

There is no heat-transfer term because no heat exchanger is present.

Step 10: Count Equations and Unknowns

Unknowns: FA, FB, FC, T

Equations: Species A Mole Balance, Species B Mole Balance, Species C Mole Balance, Energy Balance

The model is now complete.

Step 11: Apply Initial Conditions

At the reactor entrance:

FA = FA0, FB = 0, FC = 0, T = T0

using the problem statement values.

Step 12: Solve Numerically

Because:

- Multiple reactions occur simultaneously
- Rate constants depend on temperature

- Temperature depends on reaction rates

the equations are coupled ODEs.

Use Python (e.g., solve_ivp) to integrate from: V = 0 to V = 1 L

Final Answer Checklist

- [x] Calculated temperature-dependent rate constants.
- [x] Wrote separate rates for both reactions.
- [x] Created mole balances for A, B, and C.
- [x] Included both reactions in the energy balance.
- [x] Applied inlet conditions.
- [x] Solved the coupled ODE system numerically.
- [x] Generated plots of concentration and temperature versus reactor volume.

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (16) PFR Mole Balance

Eq. (44) General Rate Law

Eq. (49) Arrhenius Equation

Eq. (39) PFR Energy Balance
(Multiple Reactions)

Key Insight

Reaction Rates
↓
Determine Heat Generation
↓
Temperature Changes
↓

Rate Constants Change

↓

Reaction Rates Change

In a non-isothermal PFR with multiple reactions, temperature and reaction rates continuously influence one another throughout the reactor. The concentration and temperature profiles emerge from solving this coupled feedback system.